Multimedia Signal Processing - 2nd Module

June 19, 2023

1. **Question (7 pts)**
   Consider a discrete source X emitting the following series of symbols

   $$x = \{3,\ 15,\ 9,\ 18,\ 15,\ 15,\ -3,\ 12\}.$$

   You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

   $$Q(x) = \left\lfloor \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

   where $\lfloor \cdot \rceil$ rounds to the nearest integer, and $\Delta = 2$ is the quantization step.

   (a) Compute the sequence $\hat{x}_{\mathrm{PCM}}$ obtained by encoding and decoding x using a PCM scheme.

   (b) Based on the observation of $\hat{x}_{\mathrm{PCM}}$, compute the alphabet of the source that generated it.

   (c) Compute the Huffman code to encode $\hat{x}_{\mathrm{PCM}}$. Estimate the needed probability mass function (p.m.f.) directly from the observations you have.

   (d) Compute the sequence $\hat{x}_{\mathrm{DPCM}}$ obtained by encoding and decoding x using a closed-loop DPCM scheme. The predictor simply returns the previous sample, and the first sample is encoded using PCM.

   (e) Compare the mean squared error (MSE) obtained with the PCM and with the DPCM, commenting the achieved result.

2. **Question (5 pts)** Consider the problem of encoding an audio signal using a perceptual audio coding technique.

   (a) Describe the concept of absolute threshold of hearing, sketching it with clear labels on the axes.

   (b) Report and explain the steps implemented by the MPEG-1 psychoacoustic model 1 to determine the allowed quantization noise energy in each critical band such that the noise remains inaudible.

3. **Question (3 pts)** Consider the process of decoding an image that has been encoded using the JPEG standard.

   (a) Given a JPEG file, which pieces of information do you expect to find in the bitstream?

   (b) With the help of a block diagram, show how it is possible to decode the bitstream contained in the JPEG file into a RGB image.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider the following code containing three codewords: $C_1 = 0$, $C_2 = 1$, $C_3 = 10$. This is an instantaneous code.
   - [T] [F] Consider a discrete memoryless source X with the following alphabet $\mathcal{A} = \{\alpha, \beta, \gamma\}$. This source can have entropy $H(X) = 1$.
   - [T] [F] A Lloyd-Max scalar quantizer can improve over an Optimum Uniform scalar quantizer.
   - [T] [F] In a video coding scheme, motion vectors cannot represent motion in terms of fraction of pixels.
   - [T] [F] Consider a transform coding scheme. Usually, a single common quantizer is used to quantize every transform coefficient.
   - [T] [F] Consider a transform coding scheme that maps each input vector $\mathbf{x} = [x_1, x_2]$ into a vector of coefficients $\mathbf{y} = [y_1, y_2]$. The sum of the variances of x_1 and x_2 is equal to the sum of the variances of the transformed coefficients y_1 and y_2.
   - [T] [F] Consider two discrete sources X and Y. If the two sources has the same entropy (e.g., $H(X) = 1$ and $H(Y) = 1$), then the joint entropy is the sum of the entropies (i.e., $H(X, Y) = H(X) + H(Y)$).
   - [T] [F] Consider the problem of synthesizing speech voiced sounds using the source-filter model. If the filter is given, it is possible to synthesize the sound by feeding a periodic train of pulses to the filter.

5. **Question (Matlab - 10pts)**

   Load the build-in 'pears.png' image file using the command `img = imread('pears.png')`. Now the variable `img` contains a 3D matrix (i.e., height x width x color) representing an RGB color image. This is your input data.

   (a) Crop the top-left 256×256 pixels of the image and display it in full color (use the cropped version hereinafter).

   (b) Convert the cropped image to grayscale using any RGB-to-gray conversion of your choice (even a non-standard one).

   (c) Compute the 8x8 KLT transform matrix for the cropped grayscale image.

   (d) Apply a 2D-KLT block-wise transform encoding step using the transform obtained in the previous point and any quantizer of your choice. Pay attention to only implement the encoder, and not the decoding part.